

value of lambda used in cost-variance optimization will not be the negative value of the tangent of the strategy on the efficient trading frontier.

Additionally, some brokers specify a mapping between the strategies on the efficient trading frontier and a qualitative term. For example, using Low, Med, and High, or Passive, Normal, and Urgent, where each of these qualitative labels are mapped to different values of lambda. Other brokers may map values of lambda to be between $1 \leq \lambda \leq 10$ where 1 = passive and 10 = aggressive, or between $1 \leq \lambda \leq 3$. Some only allow values between say $0 \leq \lambda \leq 1$ in a slightly reformulated optimization such as:

$$MinL = \lambda \cdot Cost + (1 - \lambda) \cdot Risk$$

It is important to point out that there is not enough consistency in the industry to compare results based on the selected value of lambda. There are large differences across the meaning of algorithmic parameters. Investors need to understand the optimization process used by their brokers and vendors and its meaning on the cost-risk trade-off in order to make an informed trading decision.